PRAVEEN KUMAR MURALI

Robotics MSc Student | Mobile Robots and Mobile Manipulators in ROS 2, from Simulation to the Real Robot

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Portfolio

PROFILE

Erasmus Mundus master's student in Intelligent Field Robotic Systems, specialising in mobile robots and mobile manipulators in ROS 2 and in making them behave on the real robot as they did in simulation. Before the master's, worked for three years as a robotics engineer, deploying AGVs and autonomous mobile robots in factories and warehouses with ROS, ROS 2 and Nav2. Focused on building robots that work reliably and safely outside of simulation.

EDUCATION

Erasmus Mundus Joint Master in Intelligent Field Robotic Systems (IFRoS) Current
ELTE, Budapest, Hungary (3rd semester) | University of Girona, Spain (1st year)

Relevant courses: Robot Manipulation, Autonomous Systems (planning and reinforcement learning), Machine Learning, Deep Network Developments, Hands-on Intervention, Multiview Geometry

B.E. Mechatronics Engineering Aug 2019 – May 2023
Dr. Mahalingam College of Engineering and Technology, Coimbatore, India | GPA 8.9/10

PUBLICATIONS

M. Yamin, **P. K. Murali**, A. Das, H. Rajani, A. Saha, et al. *Automated Skin Lesion Detection in Total Body Photography: A Multi-Architecture Benchmark*. ISIC Workshop, MICCAI 2026 (spotlight paper).
papers.miccai.org/miccai-2026-sat/paper/ISIC_016.pdf
My part: trained the YOLO models and the RT-DETRv2 transformer detector.

ROBOTICS PROJECTS (MSc)

Terrain Quality and Hazard Mapping (ongoing) Sep 2026 – now
Intelligent Field Robots Lab, ELTE | 70 kg skid-steer field robot, ROS 2, Nav2, RealSense D435, Xsens IMU, LiDAR | team of three, my part: robot bring-up, sensors, localization, navigation, hazard fusion

- Brought up SLAM Toolbox and Nav2 on the robot: the indoor map closes its loop to 1 cm (wheels alone: 1.8 m off), and a LiDAR speed governor I wrote stops the robot 0.4 m before walls.
- Wheel slip drifted heading 15-30 degrees per run. An EKF fusing wheel speed with IMU heading (robot_localization) closed a 26 m loop to 0.7 m on recorded runs, against 4.7 m with wheels alone.
- Depth-camera hazard node (RANSAC ground plane): mean height error 1.7 mm (worst 4.3 mm) on tape-measured obstacles over 24 passes.
- Raised the IMU to 100 Hz: vibration RMS only followed speed, but the ratio of fast to slow shaking separates surfaces (asphalt about 2.5, grass 0.3). Combined camera and IMU into seen, felt and confirmed hazard classes.

Autonomous Navigation Stack, from Simulation to a Real Robot Mar – May 2026
Hands-on Planning project | ROS 2 Jazzy, Python, RTAB-Map, LiDAR | team of three

- Built an RRT* global planner and a real-time DWA local planner (10 Hz) from scratch in ROS 2, on top of RTAB-Map LiDAR SLAM, with frontier-based exploration.
- Moving from simulation to the real robot exposed clock drift between two computers (fixed with chrony), wrong scan timestamps and a LiDAR angle convention that differed from the simulator.

Pick, Transport and Place with a Mobile Manipulator Mar – May 2026
Hands-on Intervention project | ROS 2, Python, OpenCV, TurtleBot 2 + uArm | team of three

- Base and arm move together under task-priority control (damped least squares) in a 30 Hz loop, with joint limits as the top-priority task and a hysteresis band so that task doesn't switch on and off at the boundary.
- 13-state state machine for search, pick, transport and place with a suction gripper. Each step only moves on once its completion condition is met, using ArUco feedback from the camera. Tested in simulation and on the real robot.

Force-Controlled Manipulation on Industrial Arms Sep 2025 – Jan 2026
Robot Manipulation labs, University of Girona | UR10e collaborative robot (cobot, URScript, 6-axis force/torque sensor), Stäubli TX60 and TS60 (VAL3)

- UR10e polishing of a curved car hood: the tool axis runs in force mode at a constant 10 N while position and orientation stay position-controlled, so the arm follows the surface without knowing its height.

- UR10e depalletizing: the arm finds the top of a stack of unknown height by force contact, backs off 5 mm and picks with a vacuum gripper. On the Stäubli TX60, a piston-into-cylinder assembly with linear insertion, part-presence checks and recovery to a safe home position.

WORK EXPERIENCE

Research Intern | ViCOROB (Computer Vision and Robotics Institute), University of Girona Jun 2026 – Sep 2026
Supervisors: Dr. Anup Saha, Dr. Rafael Garcia

- Built a graph neural network for melanoma risk where each patient is one node and similar patients are linked, using clinical and lesion data from 472 patients.
- Evaluated performance with cross-validation, ablation studies, a simpler baseline model and bootstrap 95% confidence intervals (1,000 resamples).

Engineer, later Centre In-Charge | Forge Innovation and Ventures, India Jun 2023 – Jul 2025

- Led the technical work on 20+ robotics and vision projects for startups and industry partners, cutting their time to market by up to 30%.
- Programmed FANUC and Yaskawa robots for welding and pick and place, testing each cell and its scenarios in simulation (ROBOGUIDE, MotoSim) before deploying the programs, and fixing field faults from the robot error logs.
- Automation for Tier-1 clients (Ashok Leyland, Elkayem, Microtech), raising production-line efficiency by about 25%.
- Trained 2,000+ students and engineers in robotics and ran hackathons with 500+ participants.

Technical Intern | Forge Innovation and Ventures, Hosur, India Feb 2023 – May 2023

- Deployed a ROS-based AGV in small factories (10 kg payload, 0.5 m/s), cutting manual handling time by 40%.
- Integrated ROS 2 and the Nav2 stack with Maxbyte Technologies, which made later AGV deployments 20% faster.

Robotics Intern | Roboto AI Technologies, Coimbatore, India Aug 2022 – Jan 2023

- 2D LiDAR sensor fusion for an 80 kg warehouse robot (1 m/s), with collisions close to zero.

OTHER PROJECTS

- **Indoor and outdoor robot (2024 – 2025):** Ackermann robot with LiDAR, IMU and encoder fusion, SLAM indoors and RTK-GPS outdoors.
- **Retrofit autonomy module (2024 – 2025):** started a kit that lets existing electric tow motors drive on their own, covering sensors, control and navigation.
- **Low-cost 5-axis SCARA robot (Jul 2023 – Jan 2024):** built an 80% 3D-printed arm for pharmaceutical and food handling (0.5 kg payload, 100 mm/s, ±0.02 mm repeatability), printing the parts, assembling the mechanics and writing the code that drives the motors and sensors.
- **Wildlife monitoring with edge AI (2023 – 2024):** YOLOv8 on the device, running 24/7 for an NGO.
- **Milk quality predictor:** Random Forest model built into commercial vending machines to check milk quality in real time.

SKILLS

Robotics: ROS 2, Nav2, RViz, Gazebo, Stonefish, RTAB-Map, TurtleBot

Planning and control: RRT, RRT*, A*, DWA, frontier exploration, state machines, task-priority control

Manipulation and control: task-priority control, damped least squares, force-controlled contact search and polishing on a UR10e (URScript force mode), trajectory following, state machines with completion checks

Industrial robots: UR10e (URScript), Stäubli TX60 and TS60 (VAL3), FANUC and Yaskawa (welding, pick and place, ROBOGUIDE and MotoSim simulation), KUKA

Networked robots: multi-machine ROS 2 over Wi-Fi, QoS profiles, clock sync (chrony), communication latency, dropped messages and timestamps

Programming: Python (NumPy, pandas), C++, Git, Linux

Deep learning: PyTorch, PyTorch Geometric (graph neural networks), YOLO, RT-DETRv2, CNNs (ResNet), scikit-learn

Languages: English (fluent), Tamil (native), Kannada (fluent), Hindi (intermediate), Spanish (basic)